

Course Outline

FAST School of Management

National University of Computer & Emerging Sciences Lahore Campus



Course:	Environmental Science and Sustainability for Business		
Course code:	MG 2011	Year/Semester:	3 rd / 4 th
Program:	BBA, BS(FT), BS(BA), BS(AF)	Units/Cr Hrs:	3
Department:	FAST School of Management	Instructor(s):	Prof. Dr. Saman Shahid (PF), Ms. Shahpara Hanif (VF) & Ms. Uswa Aamir
Course Type:	Core	Email: (coordinator)	saman.shahid@lhr.nu.edu.pk
Pre-Requisite(s):	None	Phone:	
Prepared By:	Cluster Coordinator (Management)	Consultation Hours:	
Approved By:	Head of Department	Approval Date:	

COURSE DESCRIPTION

This course explores the intersection of environmental science and sustainable business practices, emphasizing the importance of sustainability in the corporate world. Students will gain a comprehensive understanding of environmental issues, climate change, environmental regulations, and ethical considerations, with a focus on how businesses can contribute to sustainable development. The objectives encompass the cultivation of a knowledgeable human resource base capable of environmental decision-making, addressing climate change issues, and facilitating business sustainability. The teaching is highly participative and includes interactive lectures. In addition, guest lectures by practitioners, experts, and researchers will be arranged to provide a holistic understanding of this course. This course will encourage students to develop their analytical and critical evaluation skills through analyzing case studies and proposing solutions. Additionally, it will enhance their team-building, management skills, and communication abilities by working on group assignments. Overall, by taking this course, students will be able to identify and understand the theoretical and practical application of an integrated environmental sciences and business sustainability approach. **The learning of this covers SDG # 8(decent work & economic growth) and SDG #9 (industry, innovation and infrastructure).**

COURSE OBJECTIVES

Upon successful completion of the course, students should be able to:		Program Educational Objectives (PEOs)
1	Acquire a holistic understanding of the environmental sustainability issues within the business context.	2,5
2	Provide a basic understanding of how businesses have engaged with the environment, considering both adverse and beneficial interactions, and assess the business strategies and solutions to tackle environmental sustainability issues.	2,3
3	Provide a basic understanding of the evolution of societal expectations regarding corporate responsibility and business sustainability, and explore sustainable business practices and corporate social responsibility.	2,5
4	Anticipate and analyze the business sustainability issues linked with the climate crisis and develop risk mitigation strategies.	2,5
5	Develop the critical, analytical, and communication skills of students through interactive and applied teaching.	1,4

COURSE LEARNING OUTCOMES (CLOs)

CLO#	Learning Outcome Statement	Program Learning Outcomes (PLOs)	Bloom's Taxonomy Level (BT)
CLO1	Understand the concepts of global environmental issues and sustainability interactions for businesses.	3,5	C2
CLO2	Explain sustainable business and corporate social responsibility initiatives that adhere to standard frameworks.	3,5	C2
CLO3	Analyze environmental business management strategies under environmental laws and regulations.	3,5,9	C3

Assessment Tool	CLO1	CLO2	CLO3
Quizzes	30	30	20
Assignments / Projects / Presentations /Class participation	20	20	30
Sessional Exam-I	50	0	0
Sessional Exam-II	0	50	0
Final Exam	0	15	35

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*Percentages are meant to be viewed vertically.



COURSE CONTENTS

CLO / Lecture #	Contents	Textbook/Course Material
Part A- Concepts on Environmental Science, Sustainability (CLO#1)		
1, #1-2	Chapters 01/24. Introducing Environmental Science and Sustainability: Human impacts on the environment, development, environment & sustainability, resource sustainable consumptions, ecological footprint, The IPAT model, global plans for sustainable future of business, sustainable development & protecting resources, sustainability & consumption, actions for sustainable future, restoring biodiversity, sustainable cities & industries, socio-economic plans, education and quality of life, HSE key performance indicators for businesses.	*Raven (textbook)
1, #3-4	Chapter 04. Ecosystem & the Physical Environment: Carbon/nitrogen/phosphorus/sulfur/hydrological cycles, temperature changes, atmospheric layers/circulation, ocean waters, weather & climate, tornadoes, tropical cyclones, hurricanes, seismic activities.	*Raven (textbook)
1, #5-6	Chapter 19. Air Pollution: Major classes/sources of air pollution (lead, PM, Noxes, Soxes, CO ₂ , hydrocarbons, etc.), ozone damage, mobile/stationary sources, urban air pollution, smog, air pollution & human health, controlling air pollution (catalytic converters, corporate average fuel economy, scrubbers), acid depositions, Clean Air Act.	*Raven (textbook)
1, #7-8	Chapter 20. Global Climate Change: Green houses gases, CO ₂ & atmosphere, modeling the future climate, extreme climate change, melting & rising sea levels, changes in precipitation patterns, effects on agriculture/health etc., international implications of global climate change, mitigation of global climate change, sequestering carbon in trees, carbon capture & storage, international efforts to reduce GHG, adaptation to global climate change, Resource Conservation & Recovery Act (RCRA), Comprehensive Environmental Response, Compensation and Liability Act (CERCLA), SUPERFUND program	*Raven (textbook)
1, #9-10	Chapter 21. Water pollution: Eutrophication, disease-causing agents, sediment problem, organic compounds, inorganic chemicals, radioactive substances, thermal pollution, water pollution from agriculture, industrial/urban/groundwater pollution, wastewater treatments, Clean Water Act	*Raven (textbook)
	Sessional I Exam	
Part B- Adopting Sustainable Strategies for Business (CLO#2)		
2, #11-12	Chapters 03/10. UN Sustainable Development Goals / Need for a Circular Economy: Coalition for Environmentally Responsible Economics (CERES), UN 2030 Agenda, the UN Sustainable Development Goals, Kalundborg symbiosis, Earlier model, designing for the circular economy.	**Nassos (textbook)
2, #13-14	Chapter 12. Adapting & Building Resilience to Climate Change: Interconnectedness of climate risks in a globalized world, large-scale climatic shifts, building resilience, case studies, and strategic development process.	**Nassos (textbook)
2, #15-16	Chapter 18. Standards & Guidelines for Managing Sustainability (Corporate Social Responsibility): Need for a sustainable strategy and standards, case studies, global reporting initiative (GRI) & stakeholders, presentation of reported stakeholder discussions (customers, communities, employees, shareholders, regulators, NGOs, suppliers), ISO 26000 framework, BSI (British Standards Institute), UN SDGs case studies (human rights, labor, environment), sustainability accounting standards board (SASB), international integrated reporting council, corporate sustainability trends.	**Nassos (textbook)
2, #17-18	Chapter 21. Sustainability Metrics: Metrics in GRI guidelines, corporate sustainability report, ecological footprint, metrics for carbon footprint, life cycle assessment (LCA), ethics and governance, environmental stewardship, stakeholder metrics & sustainability	**Nassos (textbook)
2, #19	Chapter 22. Life Cycle Analysis & Carbon Footprint Reduction: World population & energy consumption, global warming, climate change/carbon emissions/LCA/ & carbon footprint, GHG protocol, LCA & product labels, carbon offset projects.	**Nassos (textbook)
20	Guest lecture from an industrialist	
	Sessional II Exam	
Part C-Environmental Management Sustainability & Regulations (CLO#3)		
3, #21-22	Chapter 02. Environmental Economics, Laws, Policies, Regulations, & Ethics: The environmental movement events, environmental policy, federal environmental legislations, economics & the environment, strategies for pollution control, incentive based regulations, emission charge, tradable permits, marketable waste-discharge permits, corrective effect of green taxes, cost-effective analysis, environmental economics, national income accounts, cost & benefits of pollution control at the national level, environmental performance index (EPI), environmental justice, values & ethical issues, Environmental laws/regulations/policies	*Raven (textbook) † Reference material

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	from EPA USA and in Pakistan.	
3, #23-24	Chapter 03. Environmental Business Management: Understanding sustainability in businesses, Environmental Management System (EMS), Environmental Impact Assessment (EIA), Environmental Audit, Organization of environmental functions, employers and business associations, government institutions and international organizations, EIA legal requirements and guidelines, role of international organizations and development banks for EIA, case studies, Impact of sustainable technologies and digitalization on business, Environmental/ecological disruptions and business sustainability	†† Reference Book
3, #25-26	Green Business Green business: Life cycle management, resource efficiency and cleaner products, green value chains and circulatory, key concepts for green business, practice on business plan templates, green components for marketing	††† Reference Book
3, #27	Net Zero & Decarbonization: Best decarbonization pathway, organizations' digital footprints, scopes and boundaries for carbon footprinting, strategy, policies, and practices for carbon management.	†††† Reference Book
3, #28	Project Presentations and Revision	
Final Exam		

Textbooks & Reference materials

- *Environment by Peter H. Raven, David M. Hassenzahl. Wiley, 9th Edition
- **Practical Sustainability Strategies, 2nd Edition, George P. Nassos & Nikos Avlonas (Wiley, 2020)
- †Review of Environmental Laws and Regulations. By S. Asif Sial. (2018)
- ††Environmental Business Management- An Introduction. By Klaus North, 2nd Edition.
- ††Environmental Impact Assessment Handbook for Pakistan By Thomas F. Fischer
- †††Green Business Guide 2023, International Labor Organization (ILO), Italy.
- ††††The Handbook of Carbon Management: Step-by-Step Guide to High-Impact Climate Solutions for Every Manager in Every Function by Petra Molthan-Hill, Fiona Winfield, Richard Howarth, Muhammad Mazhar (2023)

Additional Reading
• Sharma, M., Luthra, S., Joshi, S. and Kumar, A. (2021), "Accelerating retail supply chain performance against pandemic disruption: adopting resilient strategies to mitigate the long-term effects", Journal of Enterprise Information Management, Vol. 34 No. 6, pp. 1844-1873. • Sahu, A., Agrawal, S. and Kumar, G. (2022), "Integrating Industry 4.0 and circular economy: a review", Journal of Enterprise Information Management, Vol. 35 No. 3, pp. 885-917. https://doi.org/10.1108/JEIM-11-2020-0465 • Valentini, C., & Kruckeberg, D. (2018). "Walking the environmental responsibility talk" in the automobile industry: An ethics case study of the Volkswagen environmental scandal. <i>Corporate Communications: An International Journal</i>, 23(4), 528-543 • https://doi.org/10.1108/CCIJ-04-2018-0045 • TEDx Talks (2024, January 03). <i>Three ways to change the Future of Climate... and everything else</i> □ EBan Goodstein □ TEDxBard College. https://www.youtube.com/watch?v=3n10SsIFB_c • Case studies that illustrate how organizations are integrating sustainability into their business models through digital transformation and analytics. https://www.avanade.com/en/solutions/sustainability/case-studies • Wagner B, Svensson G. A framework to navigate sustainability in business networks: The transformative business sustainability (TBS) model. <i>European Business Review</i>. 2014 Jun 3;26(4):340-67. • Svensson G, Wagner B. Implementing and managing economic, social and environmental efforts of business sustainability: Propositions for measurement and structural models. <i>Management of Environmental Quality: An International Journal</i>. 2015 Mar 9;26(2):195-213. • Khalife D, Subrahmanyam S, Farah A. A Sustainable Circular Business Model to Improve the Performance of Small and Medium-sized Enterprises Using Blockchain Technology. <i>Journal of Wireless Mobile Networks, Ubiquitous Computing, and Dependable Applications (JoWUA)</i>. 2024;15(2):240-50. • Sustainability in Business. 2025 https://www.ibm.com/think/topics/business-sustainability • Sustainable business practices. 2025. https://fsc.org/en/blog/sustainable-business-practices

Indigenous Material /Activity
• Asghar, A., Umer, M., & Afzal, A. (2024). Effective Implementation of Environmental Laws in Pakistan. <i>Qlantic Journal of Social Sciences and Humanities</i>, 5(1), 9-14. • Case studies from Pakistan. Videos and Documentaries • Newspaper articles from Business Recorder and Profit • Hyder S, Lussier RN. Why businesses succeed or fail: a study on small businesses in Pakistan. <i>Journal of Entrepreneurship in Emerging Economies</i>. 2016 Mar 7;8(1):82-100. • Munir T, Watts S. Exploring eco-friendly business practices and corporate innovation in Pakistan. <i>International Journal of Innovation Science</i>. 2024 Sep 20. • ESG in Pakistan. Success stories in Pakistan. 2025. https://esgsustain.secp.gov.pk/ • Balancing Economic Growth and Environmental Sustainability in Pakistan. Usman Rasheed. 2025. https://urcapk.com/economy-policy/pakistans-economic-growth-and-environmental-sustainability/



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Additional Reference Books

- Business and its Environment. David B Baron. 7th Edition, Pearson.
- Business & Sustainability by Mick Blowfield & Michael Blowfield. Oxford University Press (2013)
- Sustainability: A Comprehensive Foundation by Tom Theis and Jonathan Tomkin. (2015)
- Handbook of Sustainability-Driven Business Strategies in Practice by Markovic, Stefan; Sancha, Cristina; Lindgreen, Adam (2021)
- Sustainability: A key idea for business and society by Benn, Suzanne; Edwards, Melissa; Williams, Tima (2022)
- Absolute Essentials of Green Business. Routledge by Alan Sitkin (2020)
- Absolute Essentials of Green Business. By Alan Sitkin. Routledge Focus (2020).

ADMINISTRATIVE INSTRUCTIONS

Grading Mechanism	Weight
Announced/Surprised Quizzes	10%
Assignments	8%
Class participation	2%
Project with presentation	10%
Sessional Exam 1	10%
Sessional Exam 2	10%
Final Exam	50%

** Final Grading will be relative grading, using inter-quartile range/MCA method.

Class Policies

- Please note that any exception for one student is unfair to all other students, so don't expect any.
- Please turn off and store away cell phones, iPads, laptops, and other electronic devices.
- Talking during a lecture is not permitted. It is disrespectful and disruptive to other class members and the instructor.
- If you miss a class, it is your responsibility to determine what was covered, including any administrative announcements.

Studying

The proper way of studying for this class is following ADA policy; a short description of the same is as follows.

- Ahead of the class, it is expected that you have read the relevant chapters from the textbook.
- During the class, you are expected to follow the lecture, take notes, and ask questions; and
- After every class, you would review your notes and solve the end-of-chapter exercises, and read the textbook.

An extensive set of practice problems is placed on the Xeon Server; make sure you cover them all. We might set a tutorial session from time to time to answer your queries about the problems/practice sets.

Attendance Policy

As you can see, almost every other session is a different topic, and as such, missing any classes will result in a huge setback, as you will lose substantial course content. Besides, owing to the interactive nature of the course, attendance is crucial. Therefore, only under unavoidable circumstances, the student thinks of missing a class. *See the university policy on minimum attendance allowed for final exams.*

Scholastic Dishonesty

In the classroom and all other academic activities, students are expected to uphold the highest standards of academic integrity. Any form of scholastic dishonesty is an affront to the pursuit of knowledge and jeopardizes the quality of the degree awarded to all graduates.

What exactly is plagiarism?

- Plagiarism is a form of cheating.
- Plagiarism is using someone else's ideas or words and saying they are your own.

If you use material from a text and do not acknowledge the source, you are committing plagiarism.

Specifically, these behaviors are often regarded as plagiarism:

- Copying directly from a text, acknowledging the source but pretending that you are paraphrasing.
- Paraphrasing or copying directly from a text without acknowledging the source.
- Copying from another student's assignment with or without the student's knowledge.

The following behaviors are regarded as misconduct:

- Submitting the same assignment in two different papers.
- Getting someone else to write an assignment for you.

You are also involved in misconduct if you:

- Let another student copy your work.
- Write an assignment for another student.

Students who violate the rules on scholastic dishonesty are subject to disciplinary penalties, including the possibility of failure in the course and/or dismissal from the University. *Since dishonesty harms the individual, all students, and the integrity of the University, policies on scholastic dishonesty will be strictly and actively enforced.*